

B.2 Continuous Moments

L1

Moments of Continuous Distributions

- The defⁿ of moment, in case of discrete,

$$E[X] = \sum_x x \cdot P(X=x)$$

For continuous, it becomes -

$$E[X] = \int_x x \cdot f(x) dx$$

- Others, change accordingly -

$$E[g(x)] = \int_x g(x) \cdot f(x) dx$$

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= \int x^2 \cdot f(x) dx - \left(\int x \cdot f(x) dx \right)^2 \end{aligned}$$

Ex. A RV X has density $3x^2$ for $0 < x < 1$. Find Mean & Var.

$$E[X] = \int_0^1 x \cdot 3x^2 dx = \int_0^1 3x^3 dx = \frac{3x^4}{4} \Big|_0^1 = \frac{3}{4}$$

$$E[X^2] = \int_0^1 x^2 \cdot 3x^2 dx = \int_0^1 3x^4 dx = \frac{3x^5}{5} \Big|_0^1 = \frac{3}{5}$$

$$\text{Var} = \frac{3}{5} - \left(\frac{3}{4} \right)^2 = \frac{48}{80} - \frac{45}{80} = \frac{3}{80}$$

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Ex. If Y has density $f(y) = 1 - 0.5y$, $0 < y < 2$.
& $f(y) = 0$.

$$\begin{aligned} E[Y] &= \int_0^2 y \cdot (1 - 0.5y) dy \\ &= \int_0^2 y - 0.5y^2 dy = \frac{y^2}{2} - \frac{0.5y^3}{3} \\ &= 2 - \frac{8}{3} = \frac{6-4}{3} = \frac{2}{3} \end{aligned}$$

$$\begin{aligned} E[Y^2] &= \int_0^2 y^2 (1 - 0.5y) dy \\ &= \int_0^2 y^2 - 0.5y^3 dy = \frac{y^3}{3} - \frac{0.5y^4}{4} \\ &= \frac{8}{3} - \frac{8}{4} = \frac{32-24}{12} = \frac{8}{12} = \frac{2}{3} \end{aligned}$$

$$\begin{aligned} \text{Var}(Y) &= E[Y^2] - (E[Y])^2 \\ &= \frac{2}{3} - \frac{4}{9} = \frac{6-4}{9} = \frac{2}{9} \end{aligned}$$

L2 Moments of Mixed distributions

- In the case of mixed distributions -
1. Use discrete formula for discrete piece.

$$E[X] = \sum_x x \cdot P(X=x)$$

- 2. Use continuous formula on the continuous piece.

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

3. Then sum the two parts.

Eg. Find μ & σ^2 , if X is

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{x^2 - 2x + 2}{2} & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$

$$P(X=1) = F(1) - \lim_{x \uparrow 1} f(x)$$

$$= \frac{1}{2} - 0 = \frac{1}{2}$$

$$f(x) = \frac{d}{dx} F(x) = x - 1$$

$$E[X] = 1 \times \frac{1}{2} + \int_1^2 f(x) dx$$

$$= \frac{1}{2} + \int_1^2 x(x-1) dx$$

$$= \frac{1}{2} + \int_1^2 (x^2 - x) dx \Rightarrow \frac{x^3}{3} - \frac{x^2}{2}$$

$$= \frac{4}{3}$$

$$E[X^2] = \frac{23}{12}$$

$$\text{Var} = \frac{23}{12} - \frac{16}{9} = \frac{207 - 192}{108} = \frac{15}{108} = \frac{5}{36}$$

$$\begin{array}{r} 23 \\ \times 9 \\ \hline 207 \\ 16 \\ \times 12 \\ \hline 192 \end{array}$$

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The Survival Funcⁿ Approach

- For continuous, non-negative X ,

$$E[X] = \int_0^{\infty} P(X > x) dx$$

↳ This holds true for all non-negative RVs including discrete & mixed distributions.

$$\int_0^{\infty} P(X > x) dx = \sum_{n=0}^{\infty} P(X > n)$$

Similarly,

$$E[g(X)] = \int_0^{\infty} g'(x) P(X > x) dx$$

$$E[X^2] = \int_0^{\infty} 2x \cdot P(X > x) dx$$

$$E[X^3] = \int_0^{\infty} 3x^2 \cdot P(X > x) dx$$

- Pros -

1. often save some steps, if $F(x)$ is given but density $f(x)$ isn't.
2. faster for mixed distributions.
3. sometimes gives nicer integrals

- Cons -

1. because integral start 0, thus, it makes it messier.
2. if $f(x)$ is given, then $P(X > x)$ is extra.